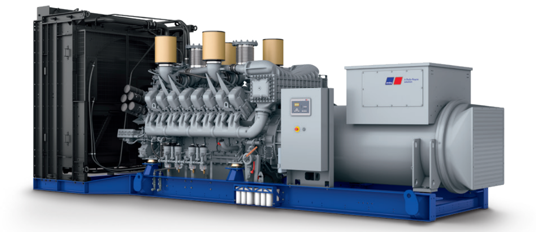
Diesel Generator Set

mtu 16V4000 DS2750

380V – 11 kV/50 Hz/standby power/fuel consumption optimized 16V4000G94F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 2720 kVA - 2870 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- Fuel consumption optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1]

Engine

Manufacturer	*mtu*
Model	16V4000G94F
Type	4-cycle
Arrangement	16V
Displacement: l	76.3
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ADEC (ECU 9)
Max power: kWm	2387
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption [2]	l/hr	g/kwh
At 100% of power rating:	567	197
At 75% of power rating:	406	188
At 50% of power rating:	277	192

Liquid capacity (lubrication)

Total oil system capacity: l	300
Engine jacket water capacity: l	175
Intercooler coolant capacity: l	50

Combustion air requirements

Combustion air volume: m^3/s	2.9
Max. air intake restriction: mbar	30

Cooling/radiator system

Coolant flow (HT-circuit) at 0,3 bar: m^3/hr	63
Coolant flow (HT-circuit) at 0,7 bar: m^3/hr	53
Coolant flow (NT-circuit) at 0,3 bar: m^3/hr	33
Coolant flow (NT-circuit) at 0,7 bar: m^3/hr	25
Heat rejection to coolant: kW	880
Heat radiated to charge air cooling: kW	590
Heat radiated to ambient: kW	90

Exhaust system

Exhaust gas temp. (after engine, max.): °C	450
Exhaust gas temp., max (after engine): °C	550
Exhaust gas temp. (before turbocharger): °C	680
Exhaust gas volume: m3/s	7.4
Maximum allowable back pressure: mbar	50

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 UL16 (Low voltage Leroy Somer standard)	380 V	2240	2800	4254	2184	2730	4148
	400 V	2240	2800	4041	2184	2730	3940
	415 V	2240	2800	3895	2184	2730	3798
Leroy Somer LSA53.2 M9 (Low voltage Leroy Somer oversized)	380 V	2240	2800	4254	2216	2770	4209
	400 V	2240	2800	4041	2216	2770	3998
	415 V	2240	2800	3895	2216	2770	3854
Leroy Somer LSA53.2 M9 (Low voltage engine output oversized)	380 V	2296	2870	4361	2216	2770	4209
	400 V	2296	2870	4142	2216	2770	3998
	415 V	2296	2870	3993	2216	2770	3854
Marathon 1020FDL7108 (Low voltage Marathon)	380 V	2240	2800	4254	2216	2720	4133
	400 V	2240	2800	4041	2216	2720	3926
	415 V	2176	2720	3784	2216	2720	3784
Leroy Somer LSA 53.2 XL11 (Med. volt. Leroy Somer)	11 kV	2280	2850	150	2208	2760	145
Leroy Somer LSA53.2 XL9 (Medium volt. Marathon)	11 kV	2280	2850	150	2208	2760	145

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.